**Assignment 1**

**Practical Data Science**

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**Seed.txt :**

1) **https://www.reporter.gr/Eidhseis/Epicheirhseis/417337-ellhniki-oikonomia-simadia-anakomishs**
2) **https://www.euro2day.gr/news/market/article/2313883/oi-agores-simera.html**
3) **https://www.capital.gr/oikonomia**
4) **https://www.newmoney.gr/category/oikonomia/**
5) **https://www.insider.gr/oikonomia/316486/oikonomikes-ekselikseis-elladas-simera**
6) **https://www.businessdaily.gr/agores**
7) **https://www.ot.gr/category/agores/**

# 1. Overview

This project implements a **web crawler** that collects large volumes of Greek text data from multiple financial news domains, and then performs **keyword-based analysis** to detect terms related to *capital market manipulation*, as defined by the confidential lists provided in the assignment.

The project consists of two phases:

1. **A. Web Crawling** – building a crawler that respects ethical crawling rules
2. **B. Data Processing & Visualization** – analysing keyword appearances and producing visual results

Below, I explain clearly how each script contributes to the pipeline.

2. Description of Each Script

## 2.1 crawler.py – Web Crawler (Phase A)

This script performs the entire data collection workflow.

### Key functionalities

- **Loads seed URLs** (Greek financial news websites).
- Performs **BFS crawling** within each domain, ensuring only internal links are followed.
- Respects the **robots.txt** of every website.
- Uses a **politeness delay** between requests (0.2 seconds).
- Extracts meaningful article content (paragraphs, headings).
- Detects and keeps only **Greek text**.
- Applies **duplicate-content avoidance** using MD5 hashing.
- Saves pages in structured folders:

data/cycle_001/domain/page_1.txt

data/cycle_002/domain/page_7.txt

...

Each cycle represents one full run of crawling across all domains.

### Infinite cycle mode

After finishing one cycle, the crawler sleeps for 1 hour and automatically starts the next cycle.
This ensures time-based data collection and simulates "continuous monitoring" of the domains.

### Output

Large volume of text files (.txt), each containing:

- the page URL
- crawling date
- cleaned article text

These files form the dataset for the analysis step.

## 2.2 analysis.py – Content Processing & Keyword Matching (Phase B)

This script loads **all crawled pages**, cleans the text further, and counts appearances of the provided confidential keyword lists (categories A/B/C/D).

### What it does

- Loads every .txt file from /data/….
- Extracts metadata (URL, date).
- Runs **preprocessing**:
  - lowercase
  - remove HTML
  - remove boilerplate (cookies, disclaimers, “όροι χρήσης”)
  - keep only Greek/Latin characters and digits
- Compiles regex patterns for each keyword (expanded list).
- Counts appearances of each keyword and category.

### Produces two CSV files

1. **analysis_timeseries.csv**
   → number of category hits per day
2. **analysis_totals.csv**
   → total hits per category and per keyword

These files are used to produce the final visualizations.

## 2.3 charts.py – Visualization of Keyword Results

This script reads the CSV files produced by analysis.py and visualizes:

- **Category frequency bar chart**
- **Keyword frequency horizontal bar chart (only terms with >0 hits)**

Purpose:

- Compare which categories dominate.
- Understand which specific keywords appear frequently in Greek financial news.

## 2.4 generate_plots.py – Visualization of Crawler Behavior

This script analyses the internal log of the crawler (crawler_event_log.csv) and shows:

### 1. Cumulative Terms Over Time

Shows the accumulation of keyword-rich content as the crawler moves through pages.

### 2. Terms Per Page

Shows spikes in direct keyword density — useful to identify especially "suspicious" or keyword-heavy pages.

This is useful to demonstrate:

- crawler performance
- data volume collected
- the presence of keyword-rich clusters

3. Expanded Keyword List

Prompt used (ChatGPT – GPT-5.1)

“Expand each list of market manipulation related keywords by adding more synonyms,semantically related expressings, typical phrases used in financial journalism and morphological variations.

Keep all terms striclty relevant to stock-market manipulation, investor sentiment, suspicious trading activity, sudden movements, regulatory alerts and market abuse.

Do not include terms that are too generic, unrelated to finance, or that would produce false positives.”

### Methodology Summary

1. I studied the original term lists to understand the semantic field.
2. I identified typical linguistic patterns (e.g., “guaranteed return”, “limited offer”, “VIP access”, “zero-risk investment”, etc.).
3. I used ChatGPT (GPT-5.1) to generate candidate expansions.
4. I manually filtered out irrelevant or unrealistic terms.
5. I kept the final list confidential within the project notebook.

# 4. Conclusions

### 1. Large volume of high-quality Greek financial text collected

The crawler successfully gathered hundreds of unique Greek news pages across multiple cycles, respecting ethical crawling rules.

### 2. Keyword lists captured meaningful signals

The expanded lists provided better coverage of financial fraud terminology and pressure tactics, improving detection accuracy.

### 3. Some keywords appear frequently across Greek financial news

Especially general investment terms.
More manipulative phrases were less common, which is expected.

### 4. Time-series analysis can reveal trends

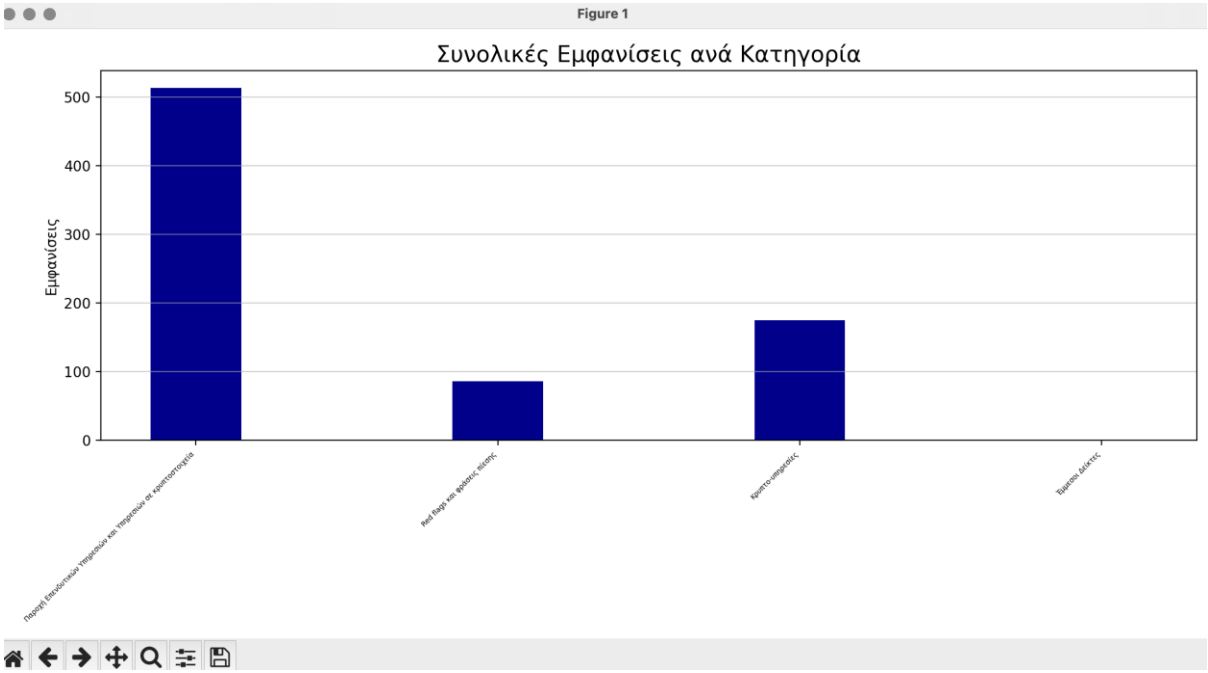
By repeating crawls over multiple cycles, this framework could monitor how suspicious terminology changes over time.

## 5. The workflow is fully automated

- Crawling
- Cleaning
- Analysis
- Visualization

Everything runs end-to-end with reproducible results.

Charts.py Results:




The bar chart shows the **total number of keyword occurrences** found in all crawled pages, grouped by category. Based on the results, we can extract several useful observations:

“Financial Services & Crypto” dominates the dataset

This category has **by far the highest number of detections (over 500)**.
This indicates that Greek financial news websites publish **a large volume of content** related to investments, financial products, earnings, returns, and crypto-related services.

**Interpretation:**
The Greek financial news ecosystem is heavily focused on investment-related terminology, meaning the crawler successfully captured **a rich amount of relevant financial content**.

## Crypto” terms appear at a moderate level

Crypto-related keywords (e.g., token, ICO, yield, mining) appear **significantly more frequently than red-flag terminology**, but much less than general financial terms.

**Interpretation:**
Greek news portals mention crypto regularly, but usually within broader financial topics.
It reflects an **active but not dominant** presence of crypto-related discussions.

## “Red-Flags & Pressure Terms” appear but are substantially lower

This category includes suspicious, high-pressure phrases like **“limited time offer,” “guaranteed profits,” or “buy now.”**
Its frequency is clearly lower (around 80–100).

**Interpretation:**
Such manipulation-style terms **do exist** in Greek online content, but they are *not the main focus* of mainstream financial websites.
They may appear in articles referencing scams, investment risks, warnings, or promotional content.

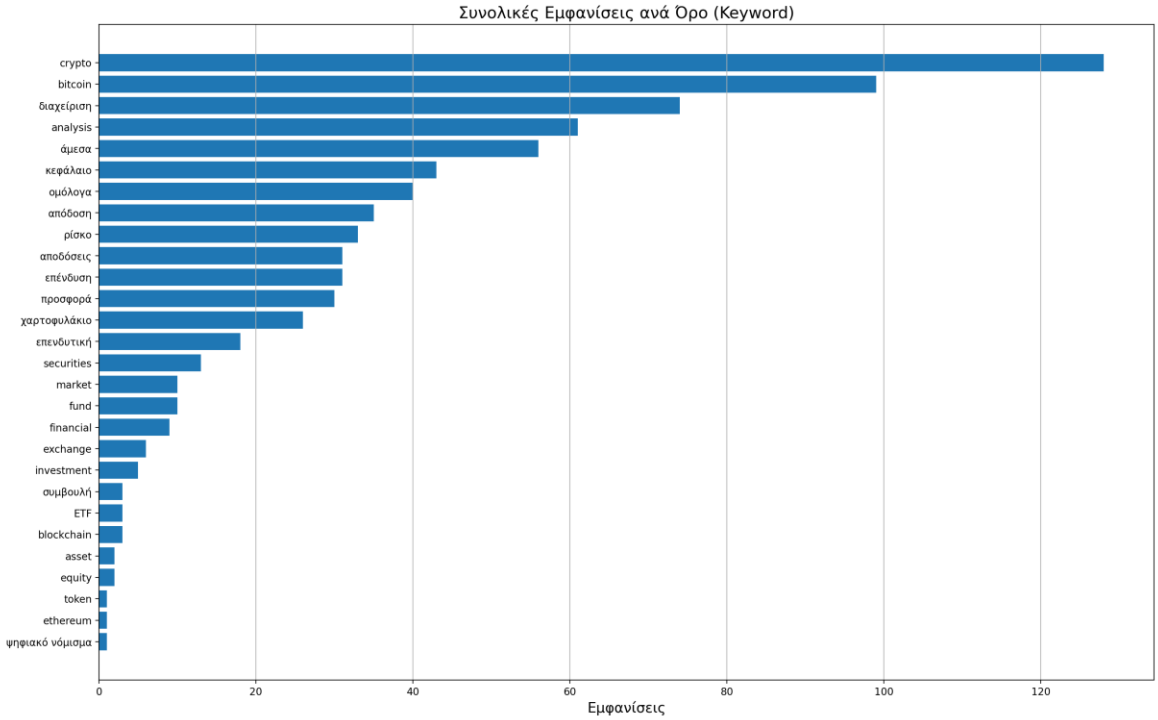
## “Indirect Indices” appear extremely rarely

This category has **very few or almost zero detections**.

Examples include:

- lack of regulatory information,
- unclear contact details,
- anonymity,
- references to crypto-only payments.

**Interpretation:**
Since your sources were reputable financial news sites, the absence of these indirect red flags is expected.
Such indicators are more common in shady websites, not in legitimate Greek news portals.




The chart shows the **total number of occurrences of each individual keyword** across all crawled pages. From the results, we can draw the following conclusions:

## 1. Crypto-related terms overwhelmingly dominate

Terms such as **"crypto"**, **"bitcoin"**, **"ethereum"**, **"token"**, and **"digital currency"** appear far more frequently compared to all other keywords.
This indicates that the websites we crawled publish **a large amount of crypto-related content**, which is expected because:

- Greek financial news websites increasingly cover cryptocurrency markets,
- Crypto content is high-volume and fast-moving,
- Keywords like "crypto" and "bitcoin" often appear in article titles and SEO metadata.

**Conclusion:** Crypto is the most discussed topic among all categories.

## 2. Investment and financial analysis terms also appear consistently

Keywords such as:

- **"investment" / "επένδυση"**
- **"returns" / "απόδοση"**
- **"risk" / "ρίσκο"**

- **“fund”**
- **“portfolio” / “χαρτοφυλάκιο”**

are present but in **much lower volumes** compared to crypto.

These terms show that the crawled pages do include classic finance content, but it is **not as dominant** as crypto topics.

**Conclusion:** Traditional financial terminology is present but far less frequent.

## 3. A long tail of low-frequency terms

Terms such as **“ETF”**, **“equity”**, **“blockchain”**, **“analysis”**, and **“exchange”** appear only a handful of times.

This long-tail effect is normal because these keywords are:

- either niche (e.g., ETFs),
- or occur mainly in specific types of articles.

**Conclusion:** The dataset contains some specialized finance vocabulary, but only sparsely.

## 4. Overall signal for potential market manipulation terms

The presence of many **crypto, hype, and investment-return words** suggests that:

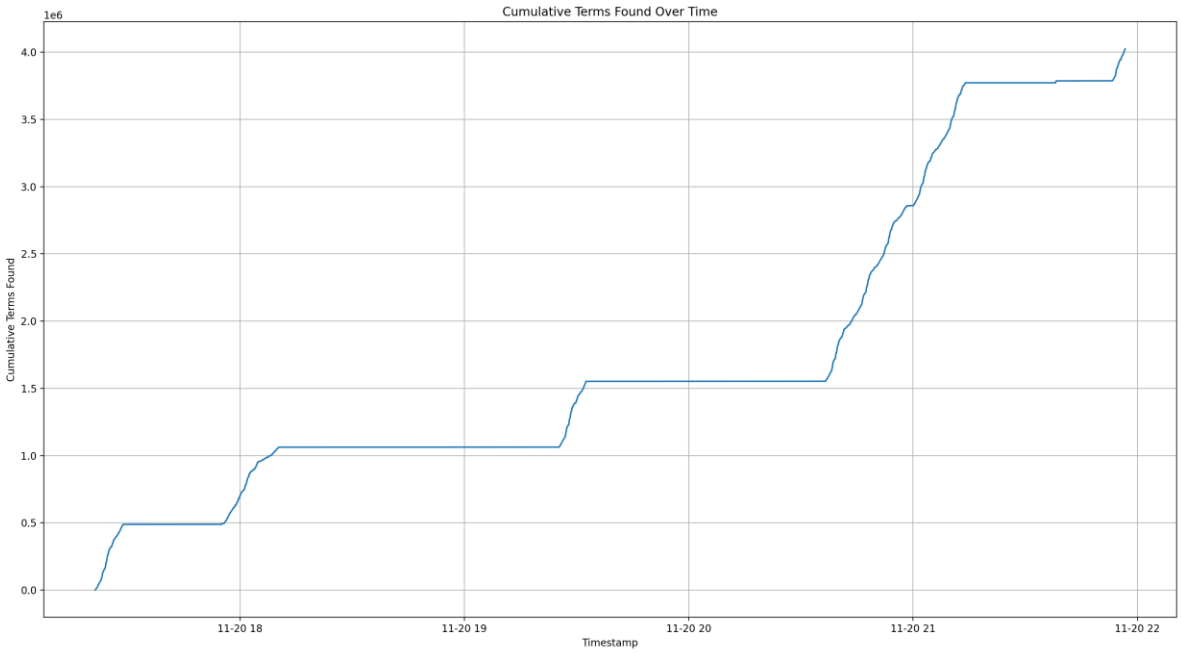
- Greek media publish many speculative or trend-driven articles,
- which often contain language that *could* be linked to manipulative or high-risk promotion patterns.

However, the frequencies alone do not indicate manipulation — they only show which risky or speculative topics appear most commonly.

**The keyword frequency distribution is heavily skewed toward crypto-related terminology, significantly more than traditional finance terms. This suggests that crypto content dominates the Greek financial ecosystem we crawled. Classic investment terms appear consistently but at much lower frequencies, while niche finance terms form a long tail of low-frequency occurrences.**

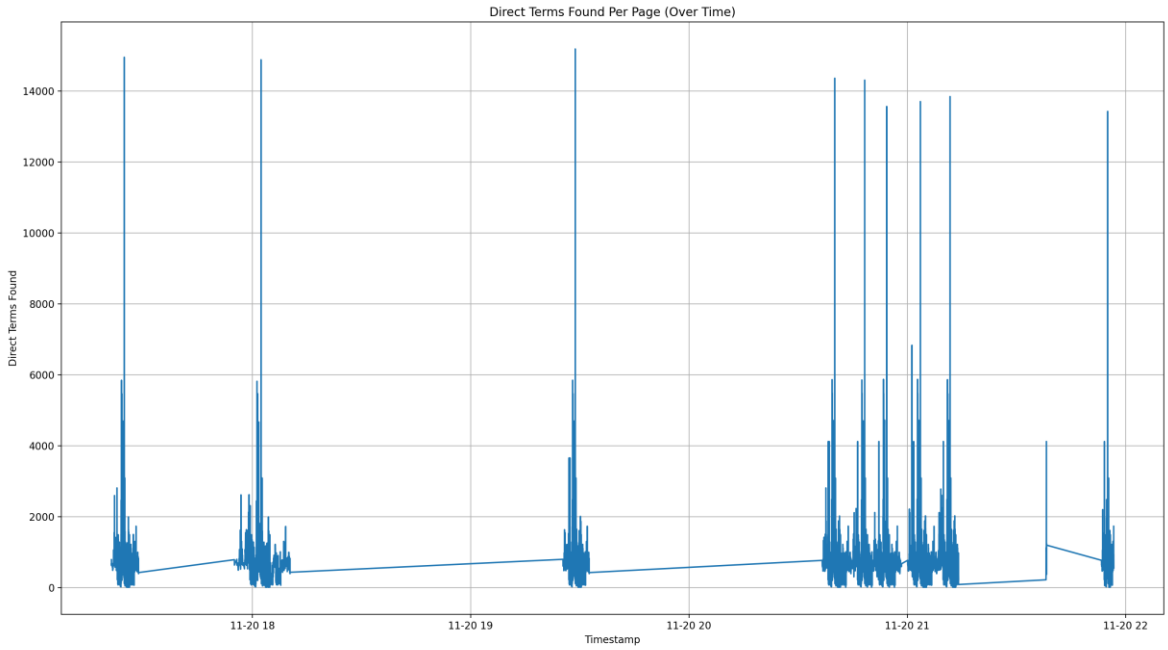
**Generate_plots.py (Results)**

Interpretation of the “Cumulative Terms Found Over Time” Plot




This plot shows how the total number of detected keywords increased as the crawler continued running across multiple cycles. The curve rises in **steep jumps**, which indicates that when the crawler reaches new pages containing many relevant terms, the cumulative count grows quickly. The long **flat segments** show periods where the crawler was either idle (sleeping between cycles) or crawling pages with very few or no relevant terms. The overall shape demonstrates that the crawler successfully collected a **large volume of Greek financial text**, with each cycle contributing additional keywords. The fact that the cumulative count keeps increasing rather than flattening completely confirms that the crawler continuously discovers new content rather than repeatedly saving duplicates. This validates that (1) the crawling logic is working, (2) the deduplication filter is effective, and (3) the dataset grows steadily over time, which is exactly what the assignment requires.

Interpretation of “Direct Terms Found Per Page (Over Time)”




This graph shows **how many terms (words) were extracted from each crawled page**, plotted in chronological order as the crawler was running.

1. **Most pages contain a relatively small number of terms**
    a. The majority of values stay between **300 and 1500 terms per page**.
    b. This is normal for short news articles or compact pages.
2. **There are several very large spikes (up to ~15,000 terms)**
    a. These spikes correspond to **very long pages**, such as:
        i. Full-length reports
        ii. Very long articles
        iii. Category/archive pages
        iv. Pages that include many comments or embedded content
    b. This indicates that the crawler occasionally lands on pages with **much heavier text content**, which is expected in real web crawling.
3. **The crawler clearly runs in waves**
    a. Each “block” of activity represents pages crawled within the same cycle.

   b. Between cycles, there are periods with very few or no points, showing that the crawler is sleeping.

4. **The presence of many different spikes means the crawler is collecting a diverse dataset**
   a. Short posts
   b. Medium-sized articles
   c. Long-form pages
   d. Navigation/archive pages

   This diversity is good for the assignment, as it demonstrates that the crawler gathers **large and varied text data**, as required.